

**T Level Technical Qualification in Digital Software Development (Level 3)****Additional Specimen Assessment Material****Time** 3 hours**Paper
reference****XXXXXX/XX****Core: Employer Set Project****TASK 1: Planning a project****Task Booklet****You must have:***Task1_RetailX_Project_Plan.xlsx***Information**

- The total mark for this set task is 18.
- This booklet is specific to each series and this material must only be issued to students who have been entered to undertake the task in the relevant series.
- This booklet must be kept securely until the start of the timetabled assessment.

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Instructions to Students

You must complete ALL parts of the activity within the assessment.

Read the assessment information carefully.

You must plan your time and submit all required evidence at the end of the supervised period.

The task must be undertaken at the time and on the date specified by Pearson.

You have **three hours** in total to complete this task.

Your provider will advise you of when any supervised breaks have been scheduled.

The task must be completed under supervised conditions.

During this task you are **not** permitted:

- to have access to the internet
- to use AI or any other tool designed to automate producing a solution.

You are permitted to use **offline** versions of relevant software to produce evidence for this task.

Files provided for use during this activity:

- *Task1_RetailX_Project_Plan.xlsx*

Your work and any material provided must be kept securely at all times.

Set Task Brief

You work for a digital systems development company that has been contracted by RetailX Ltd to plan and develop their new digital solution.

Your manager has asked you to:

- analyse the potential costs, benefits and risks associated
- produce a proposal and rationale for the development of the new digital solution.

Your manager has provided the spreadsheet file *Task1_RetailX_Project_Plan.xlsx* containing:

- a summary of the project information
- a Gantt chart template
- a staff cost calculator.

Activity

Produce a project plan which contains:

- a Gantt chart to demonstrate how you would organise the development of the new digital solution
- a plan for the resources selected and associated costs, exploring the affordability of implementing the project and the impact it will have on company profits over the next three years.

Produce a rationale that explains notable planning decisions, including:

- choices made on the order of tasks, considering any dependencies or prerequisites
- allocation of tasks to members of staff
- any potential benefits or risks you had to consider
- the impact of planning decisions on the final timings and costs of the project
- justification of the decisions you made.

Use the Set Task Information, Planning Information and project plan template provided when completing this activity.

(Total for Task 1 = 18 marks)

Outcomes for Submission

Save your project plan as an .xlsx file.

Save your rationale as a PDF file.

All files should be saved in your folder for submission.

Use this naming convention:

- *Task1_DSD_ProjectPlan_[Registration number]_[surname]_[first letter of first name]*
- *Task1_DSD_Rationale_[Registration number]_[surname]_[first letter of first name]*

Set Task Information

RetailX Ltd is a small but growing electrical goods retail chain. RetailX Ltd has a number of physical stores in the North West and also provides an online platform. The company is now expanding to other areas in the UK.

After a review of its current digital solutions, RetailX Ltd has decided that it needs to develop a new digital solution.

The aims of the new digital solution are to enable:

- staff to manage inventory and stock levels more efficiently
- management to access tools to analyse sales trends and customer behaviour
- customers to access a loyalty programme and personalised offers through a digital platform.

The requirements of the system are:

- to provide real-time inventory tracking and alerts for low stock
- to allow staff and management to interact with a dashboard of relevant sales and customer data
- to provide up-to-date information about products, promotions and customer loyalty points
- to provide customers with access through an app and website with an appropriate user interface.

Planning Information

RetailX Ltd has forecast that developing the new digital solution will lead to:

- an increase in average annual income of 7.5% after the system has been in place for one year
- an additional increase of 12% after two years
- an additional increase of 15% after three years.

The management team knows that there will be an increase in outgoing costs as a result of the new system.

RetailX Ltd and your manager have provided you with information to help you produce your plan.

Financial information for the project	
Description of the income/costs	Amount (£)
Current annual income	1 830 000
Current annual total costs	1 350 000
Forecast costs for the new digital solution	Amount (£)
Software development company base fee	60 000
Cost of training and development of staff	6 000
Cost of equipment upgrades (POS, tablets etc.)	18 585
Cost of software licensing	5 995
Cost of new server	Option 1 – hosted cloud-based server – 1 125 (per month, ongoing) Option 2 – physical server 25 600 (single payment)
Cost of infrastructure (networking) upgrades	16 000
Project manager's pay (you)	220 per day
Forecast ongoing annual costs	Amount (£)
Web hosting and mobile app store fees	5 320
Maintenance costs (server, equipment, infrastructure)	21 000
Cloud-based backup system	675 (per month)

The senior managers have allocated five members of staff to work on this project.

Your project development team will consist of:

- One senior software engineer (current employee):
 - Priya Patel
- Two junior software engineers (new employees):
 - Alex Morgan
 - Samir Khan
- One database engineer (current employee):
 - Jamie Lee
- One network engineer (current employee):
 - Morgan Evans.

Profiles for the Staff

Name	Priya Patel
Job title	Senior software engineer
Core Skills	• Expert in Python and C# • Extensive experience with Agile methodologies • Specialises in developing and integrating retail Point-of-Sale (POS) systems • Skilled in software testing and code review
Previous experience	• Eight years' experience developing solutions in the retail sector • Skilled in software testing and code review • This will be their first project as lead developer
Cost	£110 per hour

Name	Alex Morgan
Job title	Junior software engineer
Core Skills	• Specialises in full-stack solutions (HTML, CSS, JavaScript, SQL) • Good understanding of responsive design and accessibility standards
Previous experience	• 2 years' experience developing websites for small businesses • Worked as an independent web developer before joining the development company • This will be their first project with the company
Cost	£38 per hour

Name	Samir Khan
Job title	Junior software engineer
Core Skills	• Skilled in Java, SQL, Python and mobile app testing and development • Strong communication skills, both written and verbal • Able to explain technical concepts to non-technical stakeholders
Previous experience	• 3 years' experience as a mobile app developer in the health and leisure sector • Developed customer support and reward apps for gyms and fitness studios • This will be their first project with the company
Cost	£47 per hour

Name	Jamie Lee
Job title	Database engineer
Core Skills	• Specialises in SQL, Python and cloud database solutions • In-depth understanding of system security and data protection • Specialises in developing scalable back-end solutions • Good team-working and collaboration skills
Previous experience	• 7 years' experience delivering database systems for logistics and retail companies • Led migration projects from legacy systems to cloud-based platforms
Cost	£52 per hour

Name	Morgan Evans
Job title	Network engineer
Core Skills	• Excellent at problem-solving and troubleshooting network issues • Specialises in planning, implementing and monitoring network infrastructure • Specialises in analysing and improving system security • Comprehensive knowledge of network protocols and services
Previous experience	• 10 years' experience installing and maintaining networks for retail and hospitality businesses • Provided training and support for in-store staff on network best practices
Cost	£78 per hour

Your manager has produced an outline of the key tasks that need to be completed during the project. The table shows the key tasks and the estimated number of work hours it will take to complete each task.

Task	Estimated number of work hours
Upgrade infrastructure	72
Install physical server	24
Deploy and configure cloud server	20
Develop module – Stock and inventory management	108
Develop module – User interface	65
Develop module – Customer management and rewards	142
Develop module – Back-end database	120
Develop module – Sales analysis	100
Deploy modules	8 (per module)
Unit testing	21 (per module)
Integration testing	14 (per module)
Fixing and regression testing for major fault	21 (per fault found)
Fixing and regression testing for minor fault	10 (per fault found)
Deploy and configure cloud backup system	20
User/acceptance testing	35
Create a test plan	21
Training RetailX staff to use new system	34
Update website	40
Build and deploy customer rewards app	50
Hardware testing	30

It is assumed that:

- at least three minor faults will be found with each module
- no more than four major faults will be found throughout the whole project
- each member of staff will work 8 hours a day, 5 days a week
- the Database Engineer will not be available for weeks 3 and 4 of the project owing to a planned holiday.

RetailX Ltd has requested that the project is completed in **18 weeks**.

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